

1) Medical Image Enhancement

Scenario:

A doctor receives an x-ray image in which the bones are visible, but the darker tissues have very poor visibility.

1) which gray level transformation would you recommend?

Power-law (Gamma) Transformation with $\gamma < 1$.

2) why it is suitable?

It enhances the darker regions of the images without excessively affecting the brighter regions. This makes soft tissues easier to observe while keeping the bone visible.

3) what effect will it have on the image?

= * Dark tissues become brighter and more visible

* Contrast in low-intensity region increases.

* The overall image appears clearer and more informative for diagnosis.

2) Night-Time Surveillance

Scenario:

A CCTV camera captures a parking lot at night. Most pixel values are b/w 0 to 40, making the image very dark.

1) which transformation would improve the visibility?

= Log Transformation

2) why is it more suitable than negative transformation?
= Log transformation expands low-intensity (dark) pixel values, revealing detail hidden in shadows. Is negative transformation only inverts brightness and does not improve visibility in dark areas.

3) Predict the appearance of the enhanced image?

- * Dark regions become brighter.

- * Vehicles, people, and objects become easier to identify.

- * The image appears clear with improved

3) Fingerprint Recognition

Scenario:

A fingerprint image has fine ridge details, but the contrast b/w ridges and valleys is very low.

1) which gray level transformation would you apply?

= Contrast Stretching (Piecewise Linear Transformation)

2) Explain why?

= Contrast Stretching increases the difference b/w ridges and valleys, making fingerprint pattern sharper and easier to recognize by automated.

3) What would happen if a log transformation were applied instead?

A log transformation would brighten the darker pixels but would not significantly improve the contrast b/w ridges and valleys. As a result, the fingerprint details remain unclear, making recognition less accurate.